

Diagonal Polynomials and Dyadic Block Geometry in Repeated Thue–Morse Summation

A proof-oriented research note for the ProveIt Fabius-function corpus

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Abstract

Consider the two-dimensional table specified by the signed Thue–Morse prefix row and the weighted recurrence

$$s_{n,k} = \sum_{j=0}^{k-1} (k-j)s_{n-1,j}.$$

The first visible diagonals are polynomial in the row index, but the supplied code lists only eight such laws. We derive the polynomial on every diagonal, prove a complete generating-function description of the table, and explain all dyadic acceleration rules in the code.

If $r = k - n - 1 \geq 0$, then

$$s_{n,n+1+r} = D_r(n), \quad D_r(x) = \sum_{q=0}^{\lfloor r/2 \rfloor} (-1)^{\text{wt}_2(q)} \binom{2x+r-2q-1}{r-2q}.$$

Thus D_r has exact degree r , leading coefficient $2^r/r!$, is integer-valued on the half-integer lattice, and admits Stirling, Bell-polynomial, Sheffer, differential, half-step, and ruler-function recurrences. Its nonnegative half-integer zeros have an exact modular classification: $D_r(m/2) = 0$ precisely when the residue of r modulo 2^{m+1} lies in the final $m+1$ positions of the block. Negative half-integer values are finite backward differences of the signed Thue–Morse sequence.

For each fixed row n , the entire infinite sequence is a signed repetition of one nonnegative block of length 2^{2n+1} . We prove its support, circular reflection, complement law, exact central plateau, maximum, total mass, multiplicities, and number of distinct values. The same argument gives a finite-block theorem for every order of iterated summation, resolving in one factorization the periodicity and extremal patterns observed in the earlier computational question. A fully commented Python program performs exact checks and generates the accompanying profile figure.

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1 Problem, conventions, and main answer

1.1 The table

Let

$$\varepsilon_q = (-1)^{\text{wt}_2(q)}, \quad q \in \mathbb{N}_0, \quad (1.1)$$

where $\text{wt}_2(q)$ is the number of ones in the binary expansion of q . This is the signed Thue–Morse sequence. Throughout the paper, n and k are nonnegative integers. The code in the question is read as defining $s_{n,k}$ with the triangular initial condition $s_{n,k} = 0$ for $k \leq n$, the signed-prefix row at $n = 0$, and the fallback recurrence

$$s_{n,k} = \sum_{j=0}^{k-1} (k-j)s_{n-1,j}, \quad n \geq 1. \quad (1.2)$$

The symmetry branches and the first eight diagonal formulas are computational shortcuts. We will derive them rather than assume them.

The repository’s Thue–Morse atlas proves the exact prefix collapse, the iterated-prefix convolution and generating function, terminal dyadic zero runs, and signed reversal, with explicit Lean crosswalks for those statements [6]. The present note reorganizes those ingredients around the two indices (n, k) , and develops the fixed-diagonal polynomial calculus. No priority claim is made for identities that follow from standard formal power-series manipulations.

1.2 The diagonal coordinate

The first nonzero entry of row n occurs at $k = n + 1$. It is therefore natural to set

$$r = k - n - 1. \quad (1.3)$$

The visible diagonal $k - n = d$ in the original table corresponds to $r = d - 1$. We reserve $D_r(x)$ for the polynomial attached to this offset.

Closed form for an arbitrary diagonal. For every $r \geq 0$, define

$$D_r(x) = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \binom{2x + r - 2q - 1}{r - 2q} = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \frac{2x^{\overline{r-2q}}}{(r-2q)!}. \quad (1.4)$$

Then

$$s_{n,k} = \begin{cases} 0, & k \leq n, \\ D_{k-n-1}(n), & k \geq n + 1. \end{cases} \quad (1.5)$$

Equivalently, the polynomial on the original diagonal $d = k - n \geq 1$ is

$$P_d(x) = D_{d-1}(x) = \sum_{q=0}^{\lfloor (d-1)/2 \rfloor} \varepsilon_q \binom{2x + d - 2q - 2}{d - 2q - 1}. \quad (1.6)$$

It has degree $d - 1$ and leading coefficient $2^{d-1}/(d-1)!$.

Formula (1.4) is finite and exact. It is not an interpolation guessed from numerical rows: it follows coefficientwise from a single generating function, and hence applies to every row and every nonnegative column.

1.3 Summary of the structural results

The article proves the following additional facts.

1. The row generating function is

$$A_n(z) := \sum_{k \geq 0} s_{n,k} z^k = \frac{z^{n+1} E(z^2)}{(1-z)^{2n}} = \frac{z^{n+1} E(z)}{(1-z)^{2n+1}},$$

where $E(z) = \sum_{q \geq 0} \varepsilon_q z^q$.

2. The shifted row is exactly an odd-order iterated prefix:

$$s_{n,n+1+r} = T_r^{(2n+1)}.$$

Thus the diagonal polynomial interpolates a fixed coefficient as the order of summation varies.

3. The sequence $p_r(x) = r! D_r(x)$ is Sheffer for

$$E(z^2) \exp(-2x \log(1-z)).$$

In particular,

$$D_r(x) - D_r(x - \tfrac{1}{2}) = D_{r-1}(x)$$

and

$$r D_r(x) = \sum_{m=1}^r (2x + 2 - 2^{\nu_2(m)+1}) D_{r-m}(x).$$

4. For $m \geq 0$,

$$D_r(m/2) = 0 \iff r \bmod 2^{m+1} \in \{2^{m+1} - (m+1), \dots, 2^{m+1} - 1\}.$$

5. A row has block length $L = 2^{2n+1}$, block maximum $M_n = 2^{\binom{2n}{2}}$, exactly $2n+1$ zero positions, exactly $2n+1$ maximum positions, and $2^{2n} - 2n + 1$ distinct absolute values. Every nonextreme value occurs exactly twice.

2 Signed Thue–Morse prefixes and formal products

2.1 The lacunary product

The binary recursions

$$\varepsilon_{2q} = \varepsilon_q, \quad \varepsilon_{2q+1} = -\varepsilon_q \tag{2.1}$$

imply the formal product identity

$$E(z) := \sum_{q \geq 0} \varepsilon_q z^q = \prod_{j \geq 0} (1 - z^{2^j}). \tag{2.2}$$

Every coefficient of the infinite product stabilizes after finitely many factors, so (2.2) is valid in $\mathbb{Z}[[z]]$; no analytic convergence is required. Separating the first factor gives the Mahler relation

$$E(z) = (1-z)E(z^2). \tag{2.3}$$

These identities are standard in the theory of automatic sequences [1]; they are also the algebraic engine of the repository's Fabius–Rvachev development [6].

2.2 Exclusive and inclusive prefixes

Define the exclusive prefix

$$S(k) = \sum_{j=0}^{k-1} \varepsilon_j, \quad S(0) = 0. \quad (2.4)$$

Pairing $\varepsilon_{2q} + \varepsilon_{2q+1} = 0$ gives

$$S(2q) = 0, \quad S(2q+1) = \varepsilon_q. \quad (2.5)$$

Consequently,

$$\sum_{k \geq 0} S(k) z^k = zE(z^2) = \frac{zE(z)}{1-z}. \quad (2.6)$$

This is precisely the $n = 0$ row encoded by the supplied Wolfram Language branches.

For comparison with repeated summation, define inclusive iterates

$$T_r^{(0)} = \varepsilon_r, \quad T_r^{(h+1)} = \sum_{j=0}^r T_j^{(h)}. \quad (2.7)$$

One summation multiplies an ordinary generating function by $(1-z)^{-1}$, so

$$\sum_{r \geq 0} T_r^{(h)} z^r = \frac{E(z)}{(1-z)^h}. \quad (2.8)$$

For $h \geq 1$, coefficient extraction yields

$$T_r^{(h)} = \sum_{j=0}^r \binom{r-j+h-1}{h-1} \varepsilon_j. \quad (2.9)$$

Bertazzon's construction of a solution to a Thue–Morse integral equation uses these repeated sums and their dyadic block coefficients in essentially this form [2].

3 Generating function of the complete two-dimensional table

3.1 The weighted recurrence as two summations

Let

$$A_n(z) = \sum_{k \geq 0} s_{n,k} z^k. \quad (3.1)$$

The kernel in (1.2) has generating function

$$\sum_{m \geq 1} m z^m = \frac{z}{(1-z)^2}. \quad (3.2)$$

Thus (1.2) is convolution by the coefficient sequence $1, 2, 3, \dots$, or equivalently two discrete antiderivatives together with a one-step shift.

Theorem 3.1 (Exact row generating function). *For every $n \geq 0$,*

$$A_n(z) = \left(\frac{z}{(1-z)^2} \right)^n zE(z^2) = \frac{z^{n+1}E(z^2)}{(1-z)^{2n}} = \frac{z^{n+1}E(z)}{(1-z)^{2n+1}}. \quad (3.3)$$

Proof. Equation (2.6) gives $A_0(z) = zE(z^2)$. Multiplying by (3.2) implements the recurrence once, so induction gives the first two expressions. The last follows from the Mahler identity (2.3). $\square$

The triangular zero region is now automatic: A_n is divisible by z^{n+1} . More importantly, shifting by this monomial identifies each row with a standard iterated-prefix sequence.

Corollary 3.2 (Rows are odd-order iterated prefixes). *For every $n, r \geq 0$,*

$$\boxed{s_{n,n+1+r} = T_r^{(2n+1)}}. \quad (3.4)$$

Equivalently,

$$s_{n,n+1+r} = \sum_{j=0}^r \varepsilon_j \binom{2n+r-j}{2n}. \quad (3.5)$$

Proof. After removing z^{n+1} from (3.3), the remaining generating function is $E(z)/(1-z)^{2n+1}$, which is (2.8) with $h = 2n + 1$. $\square$

This relation also identifies the absolute values in the earlier question on periodic iterated Thue–Morse sums. With the notation used there,

$$|s_{n,n+1+r}| = u_r^{(2n)}. \quad (3.6)$$

The row index n therefore advances the summation order by two.

3.2 Two bivariate generating functions

Summing (3.3) over the row index gives a compact encoding of the whole array.

Proposition 3.3 (Bivariate table generating function). *As a formal power series in u with coefficients in $\mathbb{Z}[[z]]$,*

$$\boxed{\sum_{n \geq 0} \sum_{k \geq 0} s_{n,k} u^n z^k = \frac{z(1-z)^2 E(z^2)}{(1-z)^2 - uz}}. \quad (3.7)$$

For the diagonal array $D_r(n) = s_{n,n+1+r}$,

$$\boxed{\sum_{n \geq 0} \sum_{r \geq 0} D_r(n) u^n z^r = \frac{(1-z)^2 E(z^2)}{(1-z)^2 - u}}. \quad (3.8)$$

Proof. Both formulas are geometric sums. For example,

$$\sum_{n \geq 0} u^n A_n(z) = zE(z^2) \sum_{n \geq 0} \left(\frac{uz}{(1-z)^2} \right)^n.$$

Removing the shift z^{n+1} before summing gives (3.8). $\square$

4 Why fixed-index iterated sums are polynomials

The polynomial phenomenon is not peculiar to Thue–Morse. It is a universal property of repeated summation, sharpened here by the Mahler factorization.

4.1 A general summation-order polynomial

Let $a_0, a_1, \dots$ be any sequence over a characteristic-zero field and let

$$A(z) = \sum_{j \geq 0} a_j z^j.$$

For an integer $h \geq 0$, write $U_r^{(h)}$ for the h -fold inclusive sum, so its generating function is $A(z)/(1-z)^h$.

Theorem 4.1 (Polynomiality in the summation order). *For each fixed $r \geq 0$, the coefficient*

$$U_r(h) := [z^r] \frac{A(z)}{(1-z)^h} \quad (4.1)$$

extends from $h \in \mathbb{N}_0$ to the polynomial

$$U_r(h) = \sum_{j=0}^r a_j \binom{h+r-j-1}{r-j}. \quad (4.2)$$

Its degree is at most r , and is exactly r if $a_0 \neq 0$, with leading coefficient $a_0/r!$.

Proof. Expand

$$(1-z)^{-h} = \sum_{m \geq 0} \binom{h+m-1}{m} z^m$$

formally and take the Cauchy coefficient at z^r . For fixed m , the generalized binomial coefficient is a polynomial in h of degree m . Only the term $j = 0$, for which $m = r$, can contribute degree r . $\square$

The value at $h = 0$ is included: the generalized binomial vanishes for positive lower index and upper index one smaller, leaving $U_r(0) = a_r$. This polynomial continuation is therefore canonical, not an arbitrary fit.

4.2 Mahler deflation and sparse diagonals

Many automatic sequences have a functional factorization that compresses the sum (4.2).

Theorem 4.2 (A general deflation principle). *Suppose*

$$A(z) = (1-z)^c B(z^b), \quad B(z) = \sum_{q \geq 0} b_q z^q, \quad (4.3)$$

where $b \geq 2$ and $c \in \mathbb{Z}$. Then

$$U_r(h) = \sum_{q=0}^{\lfloor r/b \rfloor} b_q \binom{h-c+r-bq-1}{r-bq}. \quad (4.4)$$

Thus only one residue class modulo b of coefficient degrees occurs.

Proof. Substitute (4.3) into (4.1):

$$\frac{A(z)}{(1-z)^h} = B(z^b)(1-z)^{-(h-c)}.$$

The term $b_q z^{bq}$ leaves coefficient degree $r - bq$ in the binomial series. $\square$

For Thue–Morse, (2.3) has $b = 2$, $c = 1$, and $B = E$. Therefore

$$T_r^{(h)} = \sum_{q=0}^{\lfloor r/2 \rfloor} \varepsilon_q \binom{h+r-2q-2}{r-2q}. \quad (4.5)$$

Since the table uses $h = 2n + 1$, formula (1.4) follows immediately. This proves the main answer (1.5).

Remark 4.3 (Interpretation). For a fixed coefficient index r , $D_r(x)$ is the polynomial continuation of the h -fold signed Thue–Morse prefix under the affine change $h = 2x + 1$:

$$D_r(x) = T_r^{(2x+1)} \quad \text{in the sense of polynomial continuation in } h. \quad (4.6)$$

At integer $x = n$, this is exactly the row identity (3.4).

5 The diagonal polynomials

5.1 First examples and agreement with the supplied code

Expanding (1.4) gives the following first eight members. They are exactly the eight explicit branches in the question after replacing x by n and r by $k - n - 1$.

$$D_0(x) = 1, \quad (5.1)$$

$$D_1(x) = 2x, \quad (5.2)$$

$$D_2(x) = (x+1)(2x-1), \quad (5.3)$$

$$D_3(x) = \frac{2}{3}x(x+2)(2x-1), \quad (5.4)$$

$$D_4(x) = \frac{1}{6}(4x^4 + 12x^3 - x^2 - 3x - 6), \quad (5.5)$$

$$D_5(x) = \frac{1}{15}x(x-1)(4x^3 + 24x^2 + 39x + 34), \quad (5.6)$$

$$D_6(x) = \frac{1}{90}(x-1)(x+1)(2x-1)(4x^3 + 32x^2 + 75x + 90), \quad (5.7)$$

$$D_7(x) = \frac{1}{315}x(x-1)(x+2)(2x-1)(4x^3 + 40x^2 + 123x + 192). \quad (5.8)$$

The formula separates two kinds of structure. The rising-factorial terms control degree and coefficient growth, while the digital signs ε_q insert corrections only at degree gaps $2q$. This separation becomes transparent in the monomial coefficient formula below.

5.2 Degree, leading layers, and integrality

Theorem 5.1 (Basic shape). *For $r \geq 0$, the polynomial D_r has the following properties.*

1. $\deg D_r = r$ and

$$[x^r]D_r(x) = \frac{2^r}{r!}. \quad (5.9)$$

2. $r!D_r(x) \in \mathbb{Z}[x]$.

3. $D_r(x) \in \mathbb{Z}$ whenever $2x \in \mathbb{Z}$; in particular, every integer and half-integer value is integral.

4. At the origin,

$$D_r(0) = \begin{cases} \varepsilon_{r/2}, & r \text{ even,} \\ 0, & r \text{ odd.} \end{cases} \quad (5.10)$$

Thus every odd-indexed diagonal polynomial is divisible by x .

Proof. In (1.4), the $q = 0$ term has degree r and leading coefficient $2^r/r!$; every other term has degree at most $r - 2$. After multiplication by $r!$, each summand is an integer multiple of $2x^{\overline{r-2q}}$, which belongs to $\mathbb{Z}[x]$. If $2x = t \in \mathbb{Z}$, then $\binom{t+m-1}{m} \in \mathbb{Z}$ for every $m \geq 0$, including negative t by the generalized-binomial identity. Finally, setting $x = 0$ kills every positive rising factorial, leaving only the term with $r - 2q = 0$, when one exists. $\square$

Let $\begin{bmatrix} m \\ \ell \end{bmatrix}$ denote the unsigned Stirling number of the first kind, characterized by

$$y^{\overline{m}} = \sum_{\ell=0}^m \begin{bmatrix} m \\ \ell \end{bmatrix} y^\ell. \quad (5.11)$$

Theorem 5.2 (All monomial coefficients). *For $0 \leq \ell \leq r$,*

$$\begin{bmatrix} x^\ell \end{bmatrix} D_r(x) = 2^\ell \sum_{q=0}^{\lfloor (r-\ell)/2 \rfloor} \varepsilon_q \frac{\begin{bmatrix} r-2q \\ \ell \end{bmatrix}}{(r-2q)!}. \quad (5.12)$$

Consequently, the coefficient of x^{r-a} depends only on $\varepsilon_0, \dots, \varepsilon_{\lfloor a/2 \rfloor}$.

Proof. Insert (5.11) into the second form of (1.4) and compare powers of x . A term of degree $r - 2q$ can contribute to x^ℓ only if $q \leq (r - \ell)/2$. $\square$

The first three degree layers are especially simple. For $r \geq 3$,

$$D_r(x) = \frac{2^r}{r!} x^r + \frac{2^{r-2}}{(r-2)!} x^{r-1} + \frac{2^{r-2}(3r^2 - 7r - 22)}{24(r-2)!} x^{r-2} + \mathcal{O}(x^{r-3}). \quad (5.13)$$

For fixed r , this is also the large-row asymptotic shape of the diagonal entry $s_{n,n+1+r}$ as $n \rightarrow \infty$.

Derivation of (5.13). The leading term was already established. For degree $r - 1$, only $q = 0$ contributes, and $\begin{bmatrix} r \\ r-1 \end{bmatrix} = \begin{bmatrix} r \\ 2 \end{bmatrix}$. For degree $r - 2$, use

$$\begin{bmatrix} r \\ r-2 \end{bmatrix} = \frac{r(r-1)(r-2)(3r-1)}{24}$$

and include the leading coefficient of the $q = 1$ term, whose sign is $\varepsilon_1 = -1$. Straight simplification gives the displayed coefficient. $\square$

5.3 Finite-product and inclusion–exclusion forms

Although $E(z^2)$ is an infinite product, only finitely many factors can affect D_r . If $J = \lfloor \log_2 r \rfloor$ for $r \geq 1$, then

$$D_r(x) = [z^r] \frac{\prod_{j=1}^J (1 - z^{2^j})}{(1 - z)^{2x}}. \quad (5.14)$$

Equivalently, writing $w(S) = \sum_{j \in S} 2^j$,

$$D_r(x) = \sum_{\substack{S \subseteq \{1, \dots, J\} \\ w(S) \leq r}} (-1)^{|S|} \binom{2x + r - w(S) - 1}{r - w(S)}. \quad (5.15)$$

The unique binary representation of each even number identifies (5.15) with (1.4).

6 Sheffer structure and efficient recurrences

6.1 A Sheffer family

Define the scaled polynomials

$$p_r(x) = r!D_r(x). \quad (6.1)$$

Then the ordinary generating function in D_r is at the same time the exponential generating function in p_r :

$$\sum_{r \geq 0} \frac{p_r(x)}{r!} z^r = \sum_{r \geq 0} D_r(x) z^r = E(z^2) \exp(-2x \log(1 - z)). \quad (6.2)$$

Thus p_r is a Sheffer sequence for the pair

$$A(z) = E(z^2), \quad f(z) = -2 \log(1 - z). \quad (6.3)$$

The terminology and standard operator calculus may be found, for example, in Roman's treatment of the umbral calculus [7]. We derive every identity needed here directly from (6.2).

6.2 Half-step lowering

Theorem 6.1 (Half-step recurrence). *With the convention $D_{-1} = 0$,*

$$\boxed{D_r(x) - D_r(x - \tfrac{1}{2}) = D_{r-1}(x), \quad r \geq 0.} \quad (6.4)$$

Equivalently, for $p_r = r!D_r$,

$$(1 - e^{-\frac{1}{2}\partial_x})p_r(x) = rp_{r-1}(x). \quad (6.5)$$

Proof. Let

$$G_x(z) = \sum_{r \geq 0} D_r(x) z^r = E(z^2)(1 - z)^{-2x}.$$

Then $G_{x-1/2}(z) = (1 - z)G_x(z)$, so $G_x - G_{x-1/2} = zG_x$. Compare coefficients. $\square$

This identity is unusually useful: it says that a half-unit backward difference in the row parameter lowers the diagonal index by one. Iterating,

$$\sum_{j=0}^a (-1)^j \binom{a}{j} D_r\left(x - \frac{j}{2}\right) = D_{r-a}(x), \quad a \geq 0. \quad (6.6)$$

6.3 Differentiation and addition

Theorem 6.2 (Differential recurrence). *For $r \geq 1$,*

$$\boxed{D'_r(x) = 2 \sum_{j=1}^r \frac{D_{r-j}(x)}{j}.} \quad (6.7)$$

Proof. Differentiate $G_x(z)$ with respect to x :

$$\partial_x G_x(z) = -2 \log(1 - z) G_x(z) = 2 \left(\sum_{j \geq 1} \frac{z^j}{j} \right) G_x(z).$$

Compare coefficients. $\square$

Theorem 6.3 (Addition law). *For arbitrary x, y ,*

$$D_r(x + y) = \sum_{j=0}^r D_{r-j}(x) \binom{2y + j - 1}{j}. \quad (6.8)$$

Proof. Factor $G_{x+y}(z) = G_x(z)(1 - z)^{-2y}$ and take the coefficient of z^r . $\square$

The half-step identity is the specialization $y = -1/2$ after moving terms. The addition law also makes clear that translation in x is triangular in the diagonal index.

6.4 The ruler-function recurrence

The most efficient recurrence for generating many diagonal polynomials comes from the logarithm of G_x . Recall that $\nu_2(m)$ is the exponent of 2 in m .

Lemma 6.4 (Ruler logarithm). *One has*

$$\log G_x(z) = \sum_{m \geq 1} \frac{c_m(x)}{m} z^m, \quad c_m(x) = 2x + 2 - 2^{\nu_2(m)+1}. \quad (6.9)$$

Proof. The denominator contributes $-2x \log(1 - z) = 2x \sum_{m \geq 1} z^m/m$. On the other hand,

$$\log E(z^2) = \sum_{j \geq 1} \log(1 - z^{2^j}) = - \sum_{j \geq 1} \sum_{a \geq 1} \frac{z^{a2^j}}{a}.$$

For fixed m , the divisors $2^j \mid m$ with $j \geq 1$ contribute

$$-\frac{1}{m} \sum_{j=1}^{\nu_2(m)} 2^j = -\frac{2^{\nu_2(m)+1} - 2}{m}.$$

Adding the two parts gives (6.9). $\square$

Theorem 6.5 (Ruler recurrence). *The diagonal polynomials satisfy*

$$D_0(x) = 1, \quad r D_r(x) = \sum_{m=1}^r (2x + 2 - 2^{\nu_2(m)+1}) D_{r-m}(x). \quad (6.10)$$

Proof. Differentiate $G_x(z) = \exp(\sum c_m z^m/m)$ with respect to z :

$$z G'_x(z) = G_x(z) \sum_{m \geq 1} c_m(x) z^m.$$

Coefficient comparison gives the recurrence. $\square$

The values $2^{\nu_2(m)}$ form the classical ruler sequence. Hence all digital dependence in this polynomial recurrence enters through a one-line valuation statistic rather than explicit binary digit sums.

6.5 Bell-polynomial form

Let $\mathcal{B}_r(y_1, \dots, y_r)$ be the complete exponential Bell polynomial, defined by

$$\exp \left(\sum_{m \geq 1} y_m \frac{z^m}{m!} \right) = \sum_{r \geq 0} \mathcal{B}_r(y_1, \dots, y_r) \frac{z^r}{r!}. \quad (6.11)$$

Then (6.9) gives an immediate closed form.

Corollary 6.6 (Bell-polynomial representation).

$$D_r(x) = \frac{1}{r!} \mathcal{B}_r(0!c_1(x), 1!c_2(x), \dots, (r-1)!c_r(x)), \quad (6.12)$$

where $c_m(x) = 2x + 2 - 2^{\nu_2(m)+1}$.

Bell polynomials provide determinant, partition, and recurrence formulas through standard identities [3]. In computation, (6.10) is usually simpler, while (6.12) provides the cleanest conceptual packaging.

7 Exact half-integer roots and factor patterns

7.1 Nonnegative half-integers

The striking linear factors in the first displayed polynomials are not accidental. They are the visible part of an exact modular zero theorem.

Theorem 7.1 (Complete nonnegative half-integer zero criterion). *Let $r, m \in \mathbb{N}_0$, put*

$$h = m + 1, \quad L = 2^h. \quad (7.1)$$

Then

$$D_r(m/2) = 0 \iff r \bmod L \in \{L - h, L - h + 1, \dots, L - 1\}. \quad (7.2)$$

Equivalently,

$$(2x - m) \mid D_r(x) \text{ in } \mathbb{Q}[x] \iff r \bmod 2^{m+1} \geq 2^{m+1} - (m + 1). \quad (7.3)$$

These are exactly all zeros of D_r on $\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$.

Proof. At $x = m/2$, equation (4.6) gives

$$D_r(m/2) = T_r^{(m+1)} = T_r^{(h)}.$$

Factor the generating function at level h :

$$\begin{aligned} \frac{E(z)}{(1-z)^h} &= \frac{\prod_{j=0}^{h-1} (1 - z^{2^j})}{(1-z)^h} E(z^{2^h}) \\ &= Q_h(z) E(z^L), \end{aligned} \quad (7.4)$$

where

$$Q_h(z) = \prod_{j=0}^{h-1} (1 + z + \dots + z^{2^j-1}). \quad (7.5)$$

This polynomial has degree

$$\sum_{j=0}^{h-1} (2^j - 1) = L - h - 1$$

and every coefficient from degree 0 through $L - h - 1$ is positive. Since $E(z^L) = \sum_{q \geq 0} \varepsilon_q z^{qL}$, the coefficient in each length- L block is a signed copy of the coefficient vector of Q_h , followed by exactly h zeros. This proves (7.2). Distinct values of m give distinct roots, so the corresponding product of linear factors divides D_r . $\square$

The first residue conditions are particularly transparent:

m	root	condition on r	forced factor
0	0	$r \equiv 1 \pmod{2}$	x
1	$1/2$	$r \equiv 2, 3 \pmod{4}$	$2x - 1$
2	1	$r \equiv 5, 6, 7 \pmod{8}$	$x - 1$
3	$3/2$	$r \equiv 12, 13, 14, 15 \pmod{16}$	$2x - 3$
4	2	$r \equiv 27, 28, 29, 30, 31 \pmod{32}$	$x - 2$

For instance, D_7 must contain $x(2x - 1)(x - 1)$, exactly as in (5.8). The additional factor $x + 2$ comes from the negative half-lattice analysis below.

For fixed r , define the systematic nonnegative factor

$$\Phi_r(x) = \prod_{\substack{m \geq 0 \\ r \bmod 2^{m+1} \geq 2^{m+1} - (m+1)}} (2x - m). \quad (7.6)$$

The product is finite because $2^{m+1} - (m+1) > r$ for all sufficiently large m . The theorem says $\Phi_r \mid D_r$ in $\mathbb{Q}[x]$, up to an irrelevant nonzero scalar convention. It also completely identifies the nonnegative half-integer roots, although D_r may have additional negative rational, irrational, or complex roots.

7.2 Negative half-integers as finite differences

Extend ε_j by zero for $j < 0$, and let the backward difference be

$$(\nabla f)_r = f_r - f_{r-1}. \quad (7.7)$$

Theorem 7.2 (Negative half-lattice formula). *For every integer $a \geq 1$,*

$$D_r(-a/2) = \sum_{j=0}^{a-1} (-1)^j \binom{a-1}{j} \varepsilon_{r-j} = \nabla^{a-1} \varepsilon_r. \quad (7.8)$$

Proof. Use the alternate generating function

$$\sum_{r \geq 0} D_r(x) z^r = \frac{E(z)}{(1-z)^{2x+1}}.$$

At $x = -a/2$, this becomes $(1-z)^{a-1} E(z)$, whose coefficient is the stated backward difference. $\square$

Several consequences explain further factors in the initial list.

Corollary 7.3 (Two valuation-controlled negative roots). *For $r \geq 1$,*

$$D_r(-1) = 0 \iff \nu_2(r) \text{ is odd.} \quad (7.9)$$

For $r \geq 3$,

$$D_r(-2) = 0 \iff \nu_2(r-1) \text{ is odd.} \quad (7.10)$$

Thus $x + 1$ divides D_r under the first condition, and $x + 2$ divides it under the second.

Proof. For $x = -1$, formula (7.8) is $\varepsilon_r - \varepsilon_{r-1}$. Incrementing $r - 1$ to r changes the binary digit sum by $1 - \nu_2(r)$; hence the two signs agree exactly when $\nu_2(r)$ is odd.

For $x = -2$, the third difference is

$$\varepsilon_r - 3\varepsilon_{r-1} + 3\varepsilon_{r-2} - \varepsilon_{r-3}.$$

A four-sign check shows this vanishes exactly when $\varepsilon_r = \varepsilon_{r-3}$ and $\varepsilon_{r-1} = \varepsilon_{r-2}$. An even r cannot satisfy the second equality. If $r = 2q + 1$, both equalities reduce to $\varepsilon_q = -\varepsilon_{q-1}$, which holds exactly when $\nu_2(q)$ is even. Since $\nu_2(r - 1) = 1 + \nu_2(q)$, this is equivalent to (7.10). $\square$

For any fixed a , equation (7.8) reduces the factor question at $x = -a/2$ to a finite local pattern in the Thue–Morse word. This is an effective exact test even when a simple valuation criterion is not available.

8 Finite dyadic blocks for arbitrary summation order

This section proves the global periodic shape from which both the row acceleration rules and the half-integer root theorem arise.

8.1 The block polynomial

Fix an integer $h \geq 1$ and set

$$L = 2^h, \quad P_h(z) = \prod_{j=0}^{h-1} (1 - z^{2^j}), \quad (8.1)$$

so that $E(z) = P_h(z)E(z^L)$. Because

$$1 - z^{2^j} = (1 - z)(1 + z + \cdots + z^{2^j-1}),$$

define

$$Q_h(z) = \frac{P_h(z)}{(1 - z)^h} = \prod_{j=0}^{h-1} (1 + z + \cdots + z^{2^j-1}) = \sum_{a=0}^{L-h-1} q_{h,a} z^a. \quad (8.2)$$

The factor for $j = 0$ is simply 1. All coefficients $q_{h,a}$ in the displayed range are positive integers.

Theorem 8.1 (Complete finite-block theorem). *Let $T_r^{(h)}$ be the h -fold inclusive prefix of the signed Thue–Morse sequence. Extend $q_{h,a} = 0$ for $L - h \leq a \leq L - 1$. Then:*

1. **Signed blocks.** For $q \geq 0$ and $0 \leq a < L$,

$$\boxed{T_{qL+a}^{(h)} = \varepsilon_q q_{h,a}.} \quad (8.3)$$

Hence $|T_r^{(h)}|$ is periodic with period L .

2. **Support and terminal zeros.**

$$q_{h,a} > 0 \quad (0 \leq a \leq L - h - 1), \quad q_{h,a} = 0 \quad (L - h \leq a < L). \quad (8.4)$$

Thus every block has exactly h zero positions.

3. **Reflection.**

$$q_{h,a} = q_{h,L-h-1-a}, \quad 0 \leq a \leq L - h - 1. \quad (8.5)$$

4. **Maximum and plateau.** The maximum is

$$M_h = 2^{\binom{h-1}{2}}, \quad (8.6)$$

and it occurs exactly at

$$a = L/2 - h, L/2 - h + 1, \dots, L/2 - 1, \quad (8.7)$$

a plateau of length h .

5. **Strict ramps.** Before the plateau the positive values are strictly increasing; after it they are their reflected strictly decreasing copy.

6. **Mass.**

$$\sum_{a=0}^{L-1} q_{h,a} = 2^{\binom{h}{2}}. \quad (8.8)$$

7. **Multiplicity and distinct values.** Zero and M_h each occur exactly h times. Every other value occurs exactly twice. The number of distinct values is

$$\boxed{2^{h-1} - h + 2}. \quad (8.9)$$

8. **Complement pairing.** Put $d = L/2 - h$. For $0 \leq a < d$,

$$\boxed{q_{h,a} + q_{h,d-1-a} = M_h}. \quad (8.10)$$

Thus every value present in the block has its complement relative to M_h present as well.

Proof. From (2.8) and (8.2),

$$\frac{E(z)}{(1-z)^h} = Q_h(z)E(z^L). \quad (8.11)$$

Since $\deg Q_h = L - h - 1 < L$ and $E(z^L) = \sum_{q \geq 0} \varepsilon_q z^{qL}$, coefficient extraction immediately gives the signed-block law. The degree calculation gives the terminal zero run. The product in (8.2) has positive coefficients at every degree from zero through its degree, because it counts bounded compositions with interval-valued parts. This proves support.

Each geometric factor is reciprocal, hence Q_h is reciprocal of degree $L - h - 1$, proving reflection. To analyze the shape, split off the largest factor:

$$Q_h(z) = C_h(z)(1 + z + \dots + z^{L/2-1}), \quad C_h(z) = \prod_{j=0}^{h-2} (1 + z + \dots + z^{2^j-1}). \quad (8.12)$$

The polynomial C_h has positive coefficients on the full interval

$$0 \leq t \leq \deg C_h = L/2 - h \quad (8.13)$$

and total coefficient mass

$$C_h(1) = \prod_{j=0}^{h-2} 2^j = 2^{\binom{h-1}{2}} = M_h. \quad (8.14)$$

For $0 \leq a < L/2$, the coefficient $q_{h,a}$ is the cumulative sum of the coefficients of C_h through degree a . It therefore increases strictly until all coefficients have entered the window, at $a = L/2 - h$, and then remains equal to M_h through $a = L/2 - 1$. Reciprocity supplies the decreasing half and proves the plateau assertion.

Evaluating $Q_h(1)$ gives $\prod_{j=0}^{h-1} 2^j = 2^{\binom{h}{2}}$, proving the mass. The strict ramps and reflection show that every nonmaximum positive value occurs exactly twice. There are h terminal zeros and h maximum positions, so the left strict ramp has $L/2 - h$ entries. Counting one value per left-ramp position together with zero and the maximum gives $L/2 - h + 2$, which is (8.9).

Finally, C_h is reciprocal of degree $d = L/2 - h$. A prefix of its coefficient sequence through a and the reflected prefix through $d - 1 - a$ partition the total mass M_h , giving (8.10). $\square$

The period, block sum, maximum, number of zeros, number of maxima, distinct value count, double multiplicity, and complement property in this theorem are exactly the phenomena recorded experimentally in the 2018 question [5]. The proof shows that they are all shadows of the same finite factorization (8.11).

9 Specialization to the rows of the table

9.1 One positive block and signed repetition

For row n , set

$$h = 2n + 1, \quad L_n = 2^{2n+1}. \quad (9.1)$$

The shifted row generating function (3.3) and the block factorization give

$$A_n(z) = B_n(z)E(z^{L_n}), \quad (9.2)$$

where

$$B_n(z) = z^{n+1}Q_{2n+1}(z) = z^{n+1} \prod_{j=1}^{2n} (1 + z + \cdots + z^{2^j-1}). \quad (9.3)$$

Write

$$B_n(z) = \sum_{a=0}^{L_n-1} b_{n,a} z^a, \quad (9.4)$$

with trailing zero coefficients included to make a complete block.

Theorem 9.1 (Exact row block law). *For every $n, q \geq 0$ and $0 \leq a < L_n$,*

$$s_{n,qL_n+a} = \varepsilon_q b_{n,a}. \quad (9.5)$$

In particular, $|s_{n,k}|$ is periodic with period L_n , while the signs of consecutive blocks follow the signed Thue–Morse sequence.

Proof. This is coefficient extraction from (9.2) and $E(z^{L_n}) = \sum_{q \geq 0} \varepsilon_q z^{qL_n}$. $\square$

This proves the first large-index acceleration branch in the supplied code:

$$s_{n,k} = (-1)^{\text{TM}(\lfloor k/L_n \rfloor)} s_{n,k \bmod L_n}.$$

9.2 Complete geometry of the first block

Let

$$M_n = 2^{\binom{2n}{2}} = 2^{n(2n-1)}. \quad (9.6)$$

Shifting the arbitrary-order theorem yields the following row-specific form.

Theorem 9.2 (Geometry of row n). *For $0 \leq a < L_n$, the first block $b_{n,a}$ satisfies:*

1. **Support.**

$$b_{n,a} > 0 \iff n+1 \leq a \leq L_n - n - 1. \quad (9.7)$$

There are exactly $2n+1$ zero positions.

2. **Circular mirror.** With $b_{n,L_n} = b_{n,0} = 0$,

$$b_{n,a} = b_{n,L_n-a}, \quad 0 \leq a \leq L_n. \quad (9.8)$$

3. **Half-block complement.** For $0 \leq a \leq L_n/2$,

$$b_{n,a} + b_{n,L_n/2-a} = M_n. \quad (9.9)$$

In particular,

$$b_{n,L_n/4} = M_n/2 \quad (n \geq 1). \quad (9.10)$$

4. **Central plateau.**

$$b_{n,a} = M_n \iff L_n/2 - n \leq a \leq L_n/2 + n. \quad (9.11)$$

There are exactly $2n+1$ maximum positions.

5. **Strict ramps.** After the initial zeros, the block increases strictly to the plateau; after the plateau it decreases strictly to the final zeros.6. **Mass.**

$$\sum_{a=0}^{L_n-1} b_{n,a} = 2^{n(2n+1)}. \quad (9.12)$$

7. **Multiplicity and distinct values.** Every value other than zero and M_n occurs exactly twice. The number of distinct values is

$$2^{2n} - 2n + 1. \quad (9.13)$$

Proof. The relation between the row profile and the general block is

$$b_{n,a} = q_{2n+1,a-n-1}, \quad (9.14)$$

where $q_{h,t} = 0$ outside its natural support. Translating the support, reflection, plateau, mass, and multiplicity statements from the preceding theorem gives every assertion except the convenient form (9.9). The general complement relation translates as

$$(a - n - 1) + (L_n/2 - a - n - 1) = L_n/2 - (2n + 1) - 1,$$

which is precisely the pair of reciprocal cumulative positions in the proof of (8.10). The endpoint zero/maximum cases extend the identity to the full interval $0 \leq a \leq L_n/2$. $\square$

The mirror identity (9.8) proves the second dyadic branch in the supplied code. The complement identity (9.9) proves the third branch and strengthens its stated range from the second quarter to the entire first half. The strict lower bound in the program is necessary only to avoid a self-reference at $a = L_n/4$; equation (9.10) supplies that missing value directly.

9.3 Bounded-composition interpretation

The coefficient $b_{n,a}$ counts bounded compositions:

$$b_{n,a} = \# \left\{ (y_1, \dots, y_{2n}) : 0 \leq y_j < 2^j, \sum_{j=1}^{2n} y_j = a - n - 1 \right\}. \quad (9.15)$$

Consequently, whenever the diagonal position lies inside the first positive block,

$$D_r(n) = \# \left\{ (y_1, \dots, y_{2n}) : 0 \leq y_j < 2^j, \sum_{j=1}^{2n} y_j = r \right\}. \quad (9.16)$$

For every fixed r , this applies for all sufficiently large n . Thus the diagonal polynomial eventually counts a family of bounded compositions, while its evaluations at small n may retain signs inherited from later Thue–Morse blocks.

9.4 Normalized profiles and the Fabius connection

Figure 1 displays the normalized block

$$\frac{b_{n,a}}{M_n} \quad \text{against} \quad \frac{a}{L_n}, \quad (9.17)$$

for the first few rows. The repository develops the normalization and polygonal-interpolant limit that connects these finite coefficient grids to the Fabius function, while the same finite products become sinc products on the Fourier side and lead to the Rvachev up-function [6, 2]. The present article needs only the exact finite algebra; the figure is included to show the geometric meaning of the row identities.

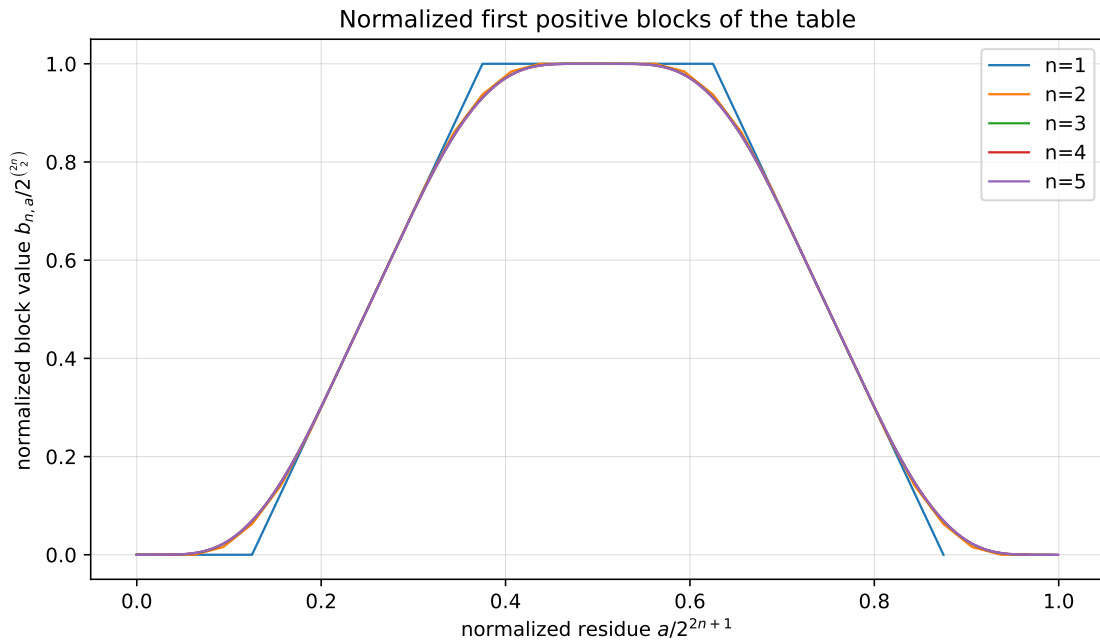


Figure 1: Normalized first positive blocks for $n = 1, \dots, 5$, generated by `experiments.py`. The central plateau has exact width $(2n + 1)/2^{2n+1}$, and the mirror/complement laws hold before taking a limit.

10 Algorithms and Wolfram Language implementation

10.1 Replacing the finite list of diagonal rules

The eight hand-written diagonal branches can be replaced by one exact definition. The following Wolfram Language code uses Pochhammer for the rising factorial and memoizes each symbolic polynomial.

Listing 1: Exact polynomial for every diagonal.

```
ClearAll[tmSign, diagonalPolynomial, x];

tmSign[q_Integer?NonNegative] := (-1)^ThueMorse[q];

diagonalPolynomial[0] = 1;
diagonalPolynomial[r_Integer?Positive] :=
  diagonalPolynomial[r] =
    Expand@Sum[
      tmSign[q] Pochhammer[2 x, r - 2 q]/Factorial[r - 2 q],
      {q, 0, Quotient[r, 2]}
    ];

ClearAll[sByDiagonal];
sByDiagonal[n_Integer?NonNegative, k_Integer?NonNegative] /; k <= n := 0;
sByDiagonal[n_Integer?NonNegative, k_Integer?NonNegative] :=
  diagonalPolynomial[k - n - 1] /. x -> n;
```

This route is ideal when $k - n$ is small and n is large: its work depends on the diagonal offset, not on the row index. A direct numeric form avoids symbolic expansion:

Listing 2: Direct exact evaluation of one diagonal entry.

```
ClearAll[sDiagonalValue];
sDiagonalValue[n_Integer?NonNegative, k_Integer?NonNegative] /; k <= n := 0;
sDiagonalValue[n_Integer?NonNegative, k_Integer?NonNegative] := Module[
  {r = k - n - 1},
  Sum[
    (-1)^ThueMorse[q] Binomial[2 n + r - 2 q - 1, r - 2 q],
    {q, 0, Quotient[r, 2]}
  ]
];
```

10.2 Generating consecutive polynomials by the ruler recurrence

When all diagonals through $r = R$ are wanted, the recurrence (6.10) reuses previous results and exposes the valuation structure.

Listing 3: Ruler-function recurrence for consecutive diagonals.

```
ClearAll[diagonalRuler, x];
diagonalRuler[0] = 1;
diagonalRuler[r_Integer?Positive] :=
  diagonalRuler[r] = Expand[
    Sum[
      (2 x + 2 - 2^(IntegerExponent[m, 2] + 1))
        diagonalRuler[r - m],
      {m, 1, r}
    ]/r
  ];
```

The recurrence is triangular and requires $R(R+1)/2$ polynomial multiplications by linear factors. For one isolated diagonal, (1.4) is usually shorter; for a full prefix of the family, the ruler recurrence is usually preferable.

10.3 Precomputing a complete row block

For fixed moderate n and many arbitrary k , precompute the one positive block B_n and use the signed block law.

Listing 4: Block-based exact table lookup.

```
ClearAll[z, rowBlockPolynomial, rowBlock, sByBlock];

rowBlockPolynomial[n_Integer?NonNegative] :=
  rowBlockPolynomial[n] = Expand[
    z^(n + 1) Product[
      Sum[z^a, {a, 0, 2^j - 1}],
      {j, 1, 2 n}
    ]
  ];

rowBlock[n_Integer?NonNegative] := rowBlock[n] = Module[
  {len = 2^(2 n + 1)},
  PadRight[CoefficientList[rowBlockPolynomial[n], z], len]
];

sByBlock[n_Integer?NonNegative, k_Integer?NonNegative] := Module[
  {len = 2^(2 n + 1), q, a},
  q = Quotient[k, len];
  a = Mod[k, len];
  (-1)^ThueMorse[q] rowBlock[n][[a + 1]]
];
```

The polynomial expansion shown here is concise and suitable for presentation. For larger n , the companion Python code computes the same coefficient list by repeated convolution with all-one windows, using a sliding sum so each factor is processed in linear time in the current output length.

10.4 Useful additional acceleration rules

The structural theorem supplies exact branches that are absent from the original listing. Inside the first block and for $n \geq 1$,

$$s_{n, L_n/4} = M_n/2, \quad (10.1)$$

$$s_{n, a} = M_n \quad \text{for } L_n/2 - n \leq a \leq L_n/2 + n, \quad (10.2)$$

$$s_{n, a} = 0 \quad \text{for } 0 \leq a \leq n \text{ or } L_n - n \leq a < L_n. \quad (10.3)$$

Together with the stronger full-half complement (9.9), these can reduce recursion depth substantially. Care is required to order conditional definitions so that fixed points such as $a = L_n/4$ are handled before a symmetry branch referring to themselves.

The standalone file `thue_morse_table.wl` packages these identities in `sFast[n, k]`. It first reduces k to its signed dyadic block, maps the local residue through the zero, plateau, reflection, and complement laws into the first quarter, and finally calls the finite diagonal formula. Thus an arbitrary entry can be evaluated without constructing the complete exponentially long block.

11 Computational verification

The archive accompanying this article contains `experiments.py`. Its identities are checked with Python integers and SymPy exact rationals; floating-point arithmetic is used only to draw Figure 1. The program performs the following independent comparisons.

1. It builds the table from the original weighted recurrence using prefix sums to evaluate each weighted convolution in constant time after preprocessing one row.
2. It evaluates $D_{k-n-1}(n)$ directly from (1.4) and compares every tested entry with the recurrence.
3. It constructs the finite block B_n by sliding-window convolutions and compares the signed-block lookup with both previous methods.
4. It mirrors the branch order of the Wolfram Language hybrid evaluator `sFast`, checks every residue of the tested row blocks, and then checks several complete signed repetitions.
5. It constructs the same diagonal polynomials independently from the ruler recurrence and verifies the half-step and differential identities symbolically.
6. It checks support, reflection, complement, plateau, total mass, multiplicities, and distinct-value counts for several rows and for arbitrary summation orders h through the tested range.
7. It verifies the nonnegative half-integer root criterion and the negative half-integer finite-difference formula, including the two valuation criteria (7.9)–(7.10).

A successful run prints

```
All exact checks passed.
```

and writes a text report of the initial factorizations. These experiments are regression checks rather than substitutes for the proofs above.

12 Further deductions and research directions

12.1 Polynomial interpolation in a discrete parameter

The variable x is not merely a formal row coordinate. Under $h = 2x + 1$, it continuously interpolates the discrete order of summation. The half-step recurrence is exactly the identity that one additional summation in h produces one lower-index coefficient after taking a difference in the order parameter. This may be useful when comparing different repeated- integration normalizations in the Fabius corpus.

12.2 Root automata and residual factors

The nonnegative half-integer roots are completely classified by a terminal block condition. At a fixed negative half-integer, the zero test is a finite linear combination of a bounded window of Thue–Morse signs. This suggests a systematic finite-automaton study of the residual linear factors of D_r as r varies. The examples $x + 1$ and $x + 2$ already reduce to elementary 2-adic valuation parity. Higher backward differences produce richer but still finite digital conditions.

12.3 Other Mahler products

The deflation theorem (4.4) applies whenever a base generating function contains a factor $(1 - z)^c$ and then contracts to z^b . For products of the form

$$E_b(z) = \prod_{j \geq 0} (1 - z^{b^j}), \quad E_b(z) = (1 - z)E_b(z^b),$$

one obtains an analogous sparse diagonal polynomial with degree gaps in multiples of b . Finite-level factorization then produces blocks of length b^h . Positivity and the exact plateau geometry require additional care when the quotient factors are no longer all-one intervals, but the polynomial mechanism survives unchanged.

12.4 Formalization targets

The repository already crosswalks Lean proofs of the exact prefix formula, iterated-prefix convolution, finite dyadic quotient, terminal zero run, and signed reversal [6]. Natural additions suggested by this article are:

1. a polynomial-valued definition of D_r and the evaluation theorem $s_{n,n+1+r} = D_r(n)$;
2. the half-step and ruler recurrences;
3. the exact half-integer zero criterion;
4. the plateau, complement, multiplicity, and distinct-value theorems for Q_h and B_n .

Each statement is finite algebra over integer polynomials once the existing product identities are imported.

13 Conclusion

The apparent list of special diagonal formulas is one family:

$$D_r(x) = [z^r] E(z^2)(1 - z)^{-2x}.$$

The Mahler equation compresses this coefficient to the explicit finite sum (1.4); the resulting polynomial has exact degree r , a controlled Stirling coefficient structure, half-lattice integrality, and a Sheffer calculus governed by the binary ruler function. Its nonnegative half-integer factors are precisely the terminal-zero positions of iterated Thue–Morse blocks.

At the global level, each row is not merely approximately periodic in shape: it is an exact signed concatenation of one finite positive block. The support, reflection, complement, plateau, maximum, mass, and multiplicity laws all follow from splitting one geometric factor from a finite Prouhet quotient. This gives both a conceptual description of the two-dimensional table and practical exact algorithms tailored to either fixed diagonals or arbitrary large columns.

A A compact theorem dictionary

For reference, the principal objects and translations are collected here.

Object	Exact formula or relation
Signed Thue–Morse	$\varepsilon_q = (-1)^{\text{wt}_2(q)}$
Lacunary series	$E(z) = \sum_q \varepsilon_q z^q = \prod_{j \geq 0} (1 - z^{2^j})$
Mahler equation	$E(z) = (1 - z)E(z^2)$
Exclusive prefix	$S(2q) = 0, S(2q + 1) = \varepsilon_q$
Iterated prefix	$\sum_r T_r^{(h)} z^r = E(z)/(1 - z)^h$
Row generating function	$A_n(z) = z^{n+1} E(z)/(1 - z)^{2n+1}$
Row/prefix map	$s_{n,n+1+r} = T_r^{(2n+1)}$
Diagonal polynomial	$D_r(x) = [z^r] E(z^2)/(1 - z)^{2x}$
Original diagonal d	$P_d(x) = D_{d-1}(x)$
Half-step	$D_r(x) - D_r(x - 1/2) = D_{r-1}(x)$
Ruler coefficient	$c_m(x) = 2x + 2 - 2^{\nu_2(m)+1}$
Ruler recurrence	$rD_r = \sum_{m=1}^r c_m D_{r-m}$
Row block length	$L_n = 2^{2n+1}$
Row maximum	$M_n = 2^{\binom{2n}{2}}$
Signed block law	$s_{n,qL_n+a} = \varepsilon_q b_{n,a}$

B Reproducibility files

The source archive contains:

- `thue_morse_diagonal_polynomials.tex`, this source;
- `thue_morse_diagonal_polynomials.pdf`, the rendered article;
- `experiments.py`, fully commented exact verification code;
- `thue_morse_table.wl`, a standalone Wolfram Language implementation of the diagonal, ruler, hybrid, and block algorithms;
- `verification_report.txt`, output from a successful run;
- `row_profiles.pdf` and `row_profiles.png`, the generated figure in vector and raster form;
- `README.txt`, build and execution instructions;
- `SHA256SUMS`, cryptographic checksums for the packaged files.

References

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